Hackney Ladish Inc.

708 South Elmira Avenue - Russellville, AR 72802 Phone: (800) 527-4500

Mill Test Report (According To: EN 10204:2004 TYPE 3.1)

Starting Material Heat #	Mfg. Heat Code	Item Description	SMLS / Welded	Material Specification/Grade	NACE Information
7503563	F17BT2C1C	16 XS CAP Y65	Seamless	MSS SP-75-2014 WPHY 65	ANSI/NACE MR0175/ISO15156:2015 ANSI/NACE MR0103/ISO17945:2015

Starting Material – Description	Plate - A572-55
Starting Material Manufacturer	
Forming Location	
Welding Location	
Heat Treatment Location	
Heat Treatment Method	Q&T 1650 1100

	Chemical Composition %															Carbon Equivalent				Hardness		Production Hardness #1		Production Hardness #2	
	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	V	Nb	Al	Ti	B	N	IIW	MSS*	PCM	CSA	BASE	WELD	BASE	WELD	BASE	WELD
Ladle																				202 BHN					
Product	.110	.180	1.290	.009	.006	.080	.090	.070	.130	.060	<.010	.0300	<.0100	<.0005		.38	.31	.20	.31						

* Applicable Carbon Equivalent calculation in accordance with the above material specification edition specified.

Impact Test																	Tensile/Yield Strength Tests							
Specimen Size	Type	Notch Orient	Temp Deg	Ft LB	Ft LB	Ft LB	Avg Ft LB	Mils Exp	Mils Exp	Mils Exp	Avg Exp	% Shear	% Shear	% Shear	Avg Shear	Loc	Base Material - Transverse .249 in. (Diameter)				Weld –			
																	Yield	Tensile	%RA	%E	Yield	Tensile	%RA	%E
																	78.5 ksi	93.8 ksi	79	28				
10mm x 10mm	T		-50 F	167	153	155	158	82	87	83	84	85	80	80	82	Base								
10mm x 10mm	T		20 F	208	181	208	199	85	79	85	83	100	85	100	95	Base								

Additional Notes

We certify that the material herein described has been manufactured in accordance with the above standards, specifications, and purchase order and satisfies all the requirements of the editions specified. We certify all materials provided comply with EN 10204:2004 type 3.1. ISO 9001 certified. The above figures are correct as contained in the records of the Company. This information has been electronically transmitted to our customer. Mark Ramey / Quality Analyst

Mark Ramey

Quality Analyst Date: 08/25/2023